Fashion-MNIST Classification

A comparison of five model architectures

Group 09 – Ryzen9

Ho Chi Minh City University of Technology
CO3133 – Deep Learning and Its Applications
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Problem and comparison goal

- One grayscale clothing image is the prediction unit.
- Input: $x \in \mathbb{R}^{1\times 28\times 28}$. Output: one of ten clothing classes.
- Compare Linear, MLP, CNN, GRU, and Transformer under a shared protocol.
- Evaluate predictive quality together with model size and runtime.

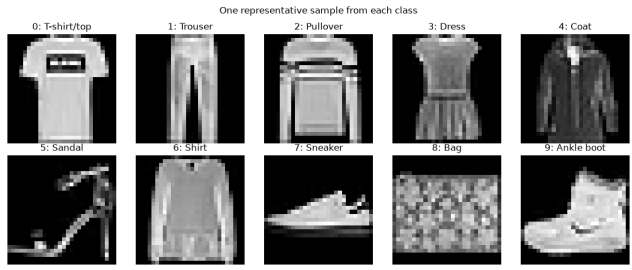
$$z = f_\theta(x) \in \mathbb{R}^{10}, \qquad \hat{y} = \arg\max_{k\in\{0,\dots,9\}} z_k$$

How does image representation affect quality and computational cost?

Fashion-MNIST and exploratory analysis

- 70,000 labeled images.
- 28×28 pixels, one grayscale channel.
- Ten categories.
- 6,000 images per class in the official training set.

No class weighting or resampling.



Balanced counts still leave visually difficult categories.

Official Fashion-MNIST dataset. Original training examples from the EDA.

Shared split and preprocessing

Partition	Images	Per class
Training	54,000	5,400
Validation	6,000	600
Test	10,000	1,000

Stratified split, seed 42.

Shared partitions for all five models.

- Scale pixels to $[0, 1]$.
- Estimate statistics from training images only.
- Keep the original image size and labels.
- No data augmentation.

$$\tilde{x} = \frac{x - 0.286209}{0.353160}$$

Validation selects the checkpoint before held-out test evaluation.

Model input representations

Model	Representation	Shape per image
Linear / MLP	Flattened pixels	784
CNN	Spatial feature maps	$1 \times 28 \times 28$ input
GRU	Sequence of image rows	28 steps, 28 features
Transformer	4×4 patches	49 patch tokens

Representation changes which relationships a model can learn easily.

Linear classifier and MLP

Linear

$$z = Wv + b, \qquad v \in \mathbb{R}^{784}$$

- One affine layer with ten outputs.
- 7,850 parameters.
- Compact reference model.
- Limited to a linear decision boundary.

MLP

$$784 \longrightarrow 256 \longrightarrow 128 \longrightarrow 10$$

- ReLU after each hidden layer.
- Dropout 0.2 after the second ReLU.
- 235,146 parameters.
- Nonlinear combinations of flattened pixels.

Both output raw logits for cross-entropy loss.

CNN architecture

- Two blocks: 3×3 convolution, batch normalization, ReLU, max pooling, and spatial dropout.
- Convolution shares local filters across image positions.
- Pooling reduces feature-map resolution.

$$1 \times 28 \times 28 \longrightarrow 32 \times 14 \times 14 \longrightarrow 64 \times 7 \times 7$$

Adaptive pooling to $64 \times 2 \times 2 \longrightarrow 256 \longrightarrow 128 \longrightarrow 10$

53,098 parameters with an explicit preference for local image structure.

GRU and Transformer

GRU

- Read 28 image rows in sequence.
- 28 input features per timestep.
- One layer, 128 hidden units.
- Final hidden state feeds ten logits.

$$h_t = \mathrm{GRU}(x_t, h_{t-1})$$

61,962 parameters.

Transformer

- 49 patch tokens plus a class token.
- 128-dimensional embeddings and positions.
- Four layers, four attention heads.
- Class token feeds ten logits.

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{32}}\right)V$$

803,338 parameters.

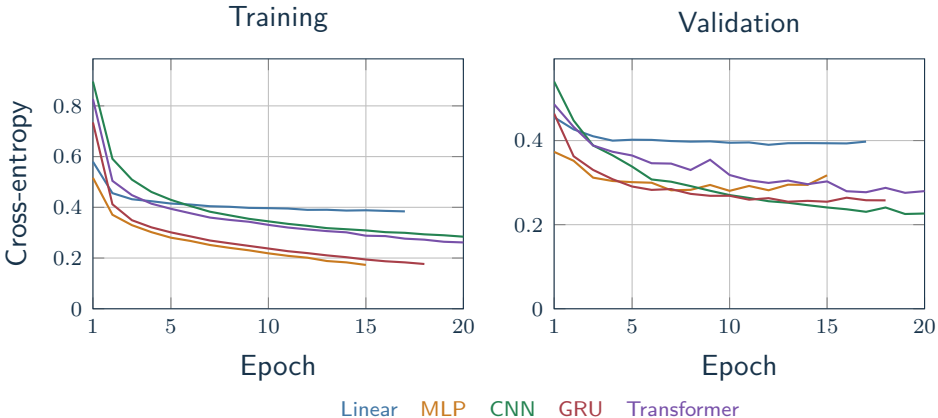
Training and experimental protocol

Loss	Cross-entropy
Optimizer	AdamW
Learning rate	0.001, fixed
Weight decay	0.0001
Batch size	128
Epoch limit	20
Seed	42

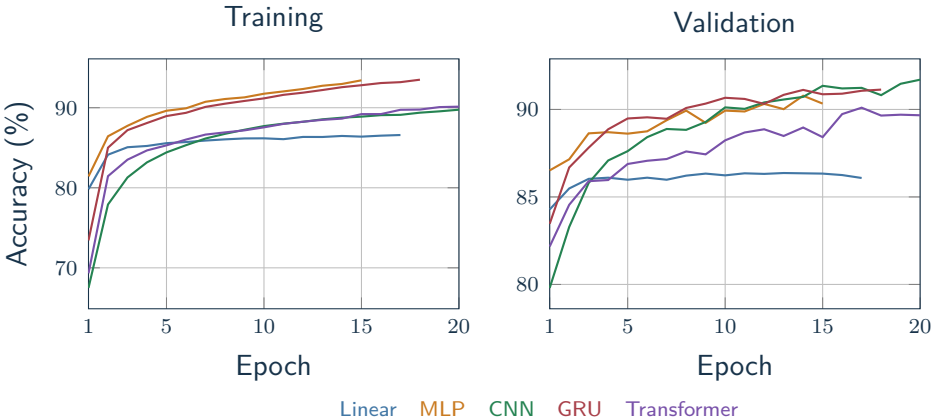
- Early stopping: patience 5, minimum improvement 0.0001.
- Checkpoint: lowest observed validation loss.
- One baseline run per model.
- Shared metrics and test protocol.

NVIDIA GeForce RTX 5070 Ti. FP32 training. No scheduler or mixed precision.

Training and validation loss



Training and validation accuracy



Held-out test performance

Model	Test loss	Accuracy (%)	Macro-F1
Linear	0.4455	84.23	0.8395
MLP	0.3318	88.56	0.8855
CNN	**0.2691**	**90.23**	**0.9023**
GRU	0.2926	89.72	0.8970
Transformer	0.3015	89.16	0.8908

CNN leads these recorded runs on both accuracy and Macro-F1.

10,000 test images. Each model uses its best-validation-loss checkpoint.

Model size and computational cost

Model	Parameters	Epochs	Training (s)	Inference (ms/image)
Linear	7,850	17	290.77	0.0013
MLP	235,146	15	257.38	0.0022
CNN	53,098	20	343.46	0.0070
GRU	61,962	18	305.00	0.0033
Transformer	803,338	20	362.18	0.0232

Linear is smallest and fastest. GRU trades some accuracy for lower inference cost than CNN.

Inference measures batched forward passes. Training includes validation and checkpoint saving.

Upper-body categories remain difficult

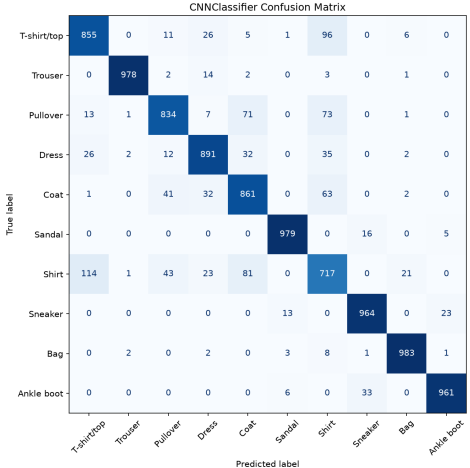
Recall (%)	Linear	MLP	CNN	GRU	Transformer
T-shirt/top	81.8	85.9	85.5	82.9	87.2
Pullover	75.3	75.7	83.4	79.1	88.5
Coat	75.0	87.5	86.1	86.2	81.1
Shirt	50.9	68.7	71.7	71.8	64.8

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

The overall winner does not lead every category.

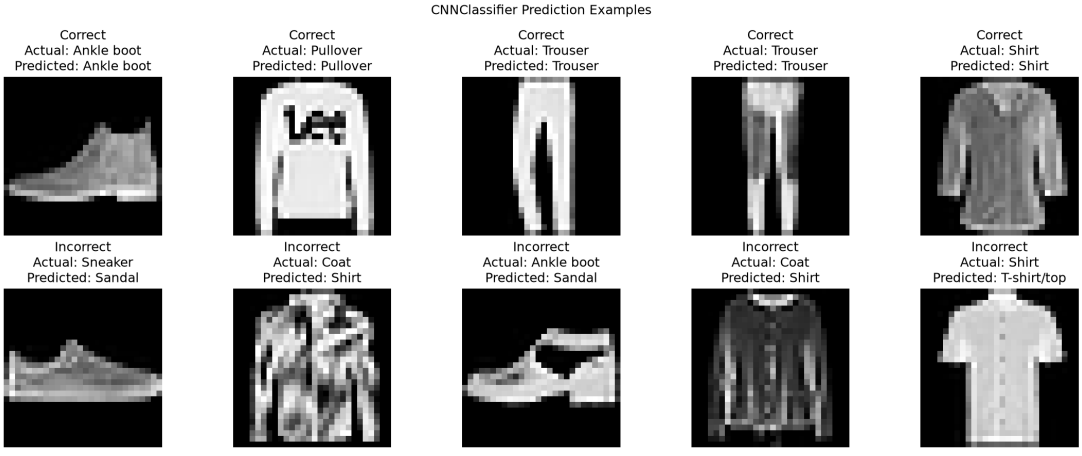
Each test class contains 1,000 images.

CNN confusion matrix



Rows: true class. Columns: predicted class. CNN makes 977 test errors.

Correct and incorrect predictions



First five correct and first five incorrect CNN predictions in test-set order.

Representation and quality-cost trade-offs

- MLP gains 4.33 accuracy points over Linear through nonlinear features, with a much larger parameter count.
- CNN gains 1.67 points over MLP with about 77.4% fewer parameters.
- GRU is 0.51 points below CNN with about 52.9% less inference time.
- Transformer has about 15.1 times CNN's parameters and lower accuracy.

The observed CNN advantage is consistent with useful spatial structure.

Architecture, width, and regularization vary together across these runs.

Limitations and conclusion

- One run per model at seed 42 leaves seed variability unmeasured.
- Shared settings do not individually optimize each architecture.
- Different capacities and epoch counts limit causal comparisons.
- Small grayscale product images limit generalization to retail scenes.

CNN gives the strongest observed predictive result. Linear and GRU offer useful efficiency trade-offs.

Next experiments: repeat seeds, tune on validation data, and test one controlled ablation.

Reproduction and short demonstration

Run from the `assignment1` directory using the released code and configuration.

```
# Train and evaluate a new baseline
python -m experiments.run --model linear --epochs 20 --run-name demo --evaluate-test

# Evaluate the downloaded baseline checkpoint
python -m experiments.run --model linear --evaluate-only --run-id 20260924-124837-513213-m2-
    baseline
```

- Baseline source tag: a1-m2-baselines-2026-09-25
- Checkpoint archive and recorded run artifacts

AI assistance and sources

- Codex (OpenAI) assisted with slide text, LaTeX structure, and speaker-note drafts.
- Reported metrics originate from executed model runs.
- The group is responsible for final review and explaining the work.

Sources: course handbook Sections 2.3 and 12, the Assignment 1 report, saved baseline logs, and official Fashion-MNIST documentation.